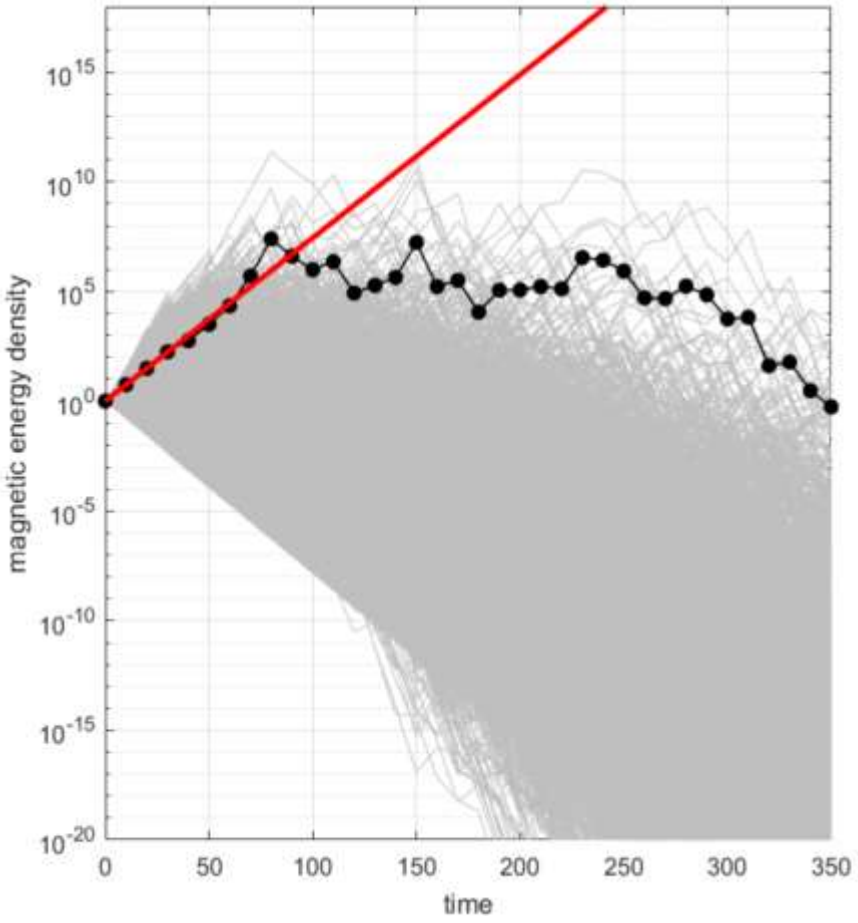
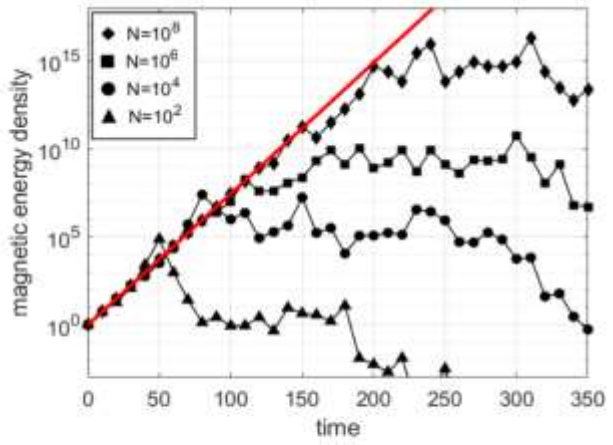
Стохастика, искусственный интеллект и финансовая математика
ВШЭ, 14 октября 2026

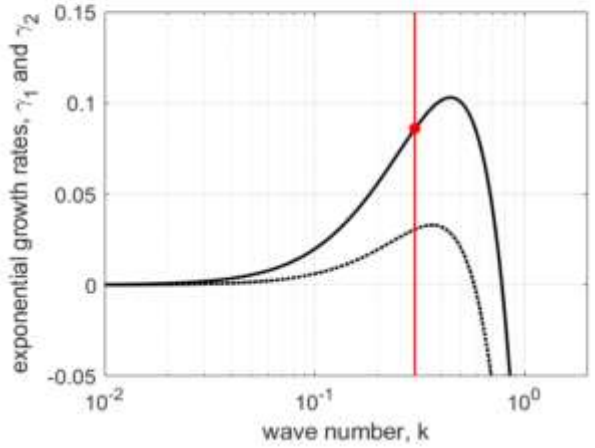
Dynamics in the intermittent media

Egor ILLARIONOV, *MSU*

Numerical modelling of intermittency







Yushkov, Illarionov, Sokoloff. Helicity fluctuations in the mean-field dynamo model. Chaos (under review). 2026

Illustrative example

$$\partial_t \mathbf{B} = \alpha \nabla \times \mathbf{B} + \beta \Delta \mathbf{B} \quad \Longrightarrow \quad \partial_t \hat{\mathbf{B}} = -(i\alpha L + \beta k^2 I)\hat{\mathbf{B}}$$

Fourier transform

Solution for constant α:

$$\hat{\mathbf{B}} = \exp(-\beta k^2 t)\left(\cosh(\alpha k t)\hat{\mathbf{B}}_0 + i \sinh(\alpha k t)[\mathbf{k}/k, \hat{\mathbf{B}}_0]\right)$$

Higher modes ($k < |\alpha|/\beta$) grow exponentially,
all modes decay when $\alpha = 0$

Possibility of growth at $\langle \alpha \rangle = 0$?

Random α with finite memory time

$$\hat{\mathbf{B}}(\tau(n+1)) = \exp(-\beta k^2 \tau) \exp(-i\alpha_n \tau L) \hat{\mathbf{B}}(\tau n)$$

The dynamics is governed by the product of stochastic matrix exponentials

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \ln ||C_n C_{n-1} \dots C_1 \hat{\mathbf{B}}_0||$$

Growth rate of individual realization (Lyapunov exponent)

$$\lambda = \langle\alpha\rangle k\tau - \beta k^2$$

$\langle\alpha\rangle = 0$ => with probability 1 each realization decays sooner or later

Random α with finite memory time

$$\langle \hat{\mathbf{B}}(\tau(n+1)) \rangle = \exp(-\beta k^2 \tau) \langle \exp(-i\alpha_n \tau L) \rangle \langle \hat{\mathbf{B}}(\tau n) \rangle$$

The dynamics of the mean field is defined by the eigenvalues of the averaged matrix exponential

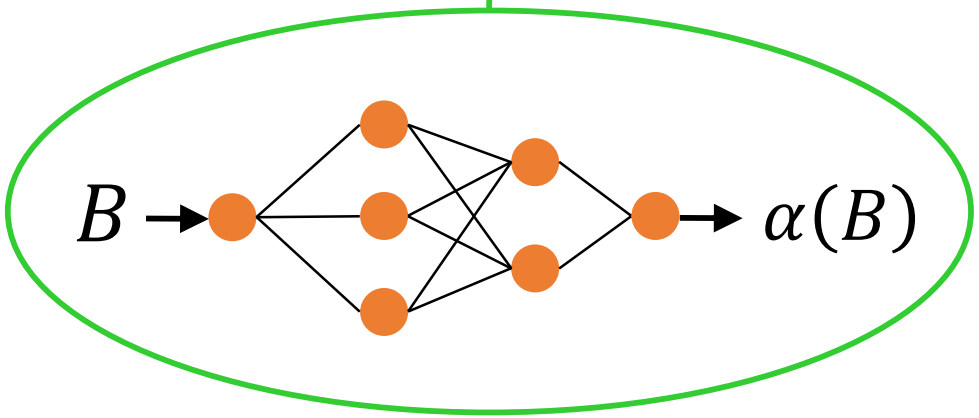
Growth rate of the mean field: $\gamma_1 = \frac{1}{\tau} \ln \langle \cosh(\alpha k \tau) \rangle - \beta k^2$

$$\alpha \sim N(0, \sigma^2) \Rightarrow \gamma_1 = \frac{1}{2} \sigma^2 k^2 \tau - \beta k^2$$

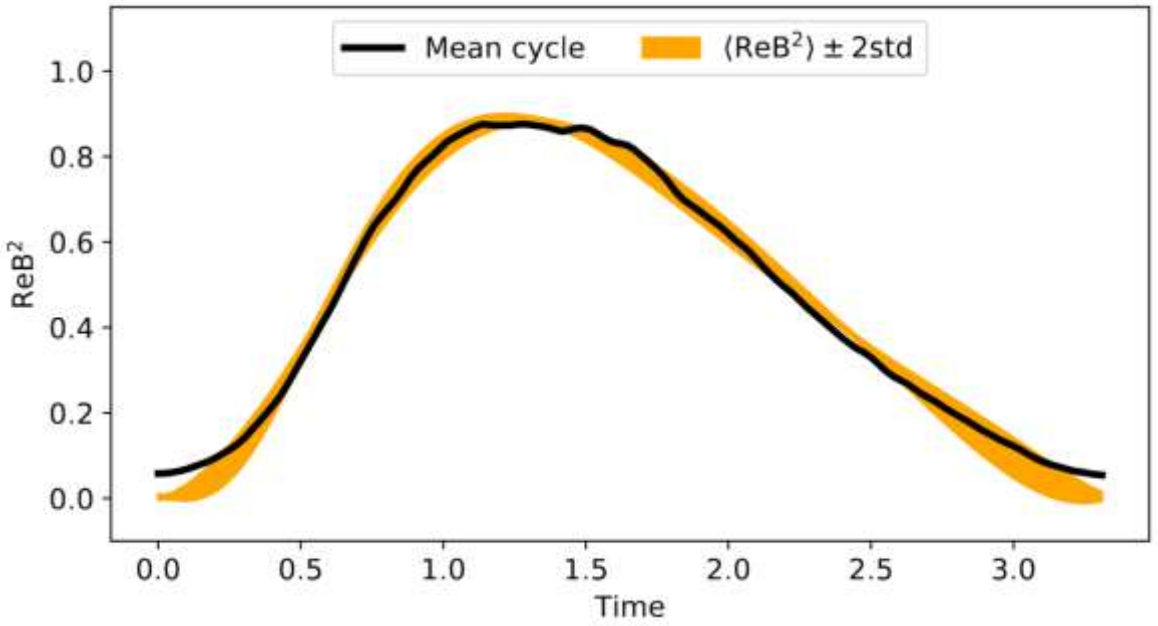
Mean field grows exponentially when τ is sufficiently large

Neural-differential dynamo models for data assimilation

$$\begin{cases} \dot{B}(t) = -iDA(t) - B(t) \\ \dot{A}(t) = \alpha\big(B(t)\big)B(t) - A(t) \end{cases}$$



Unknown function are replaces with neural networks

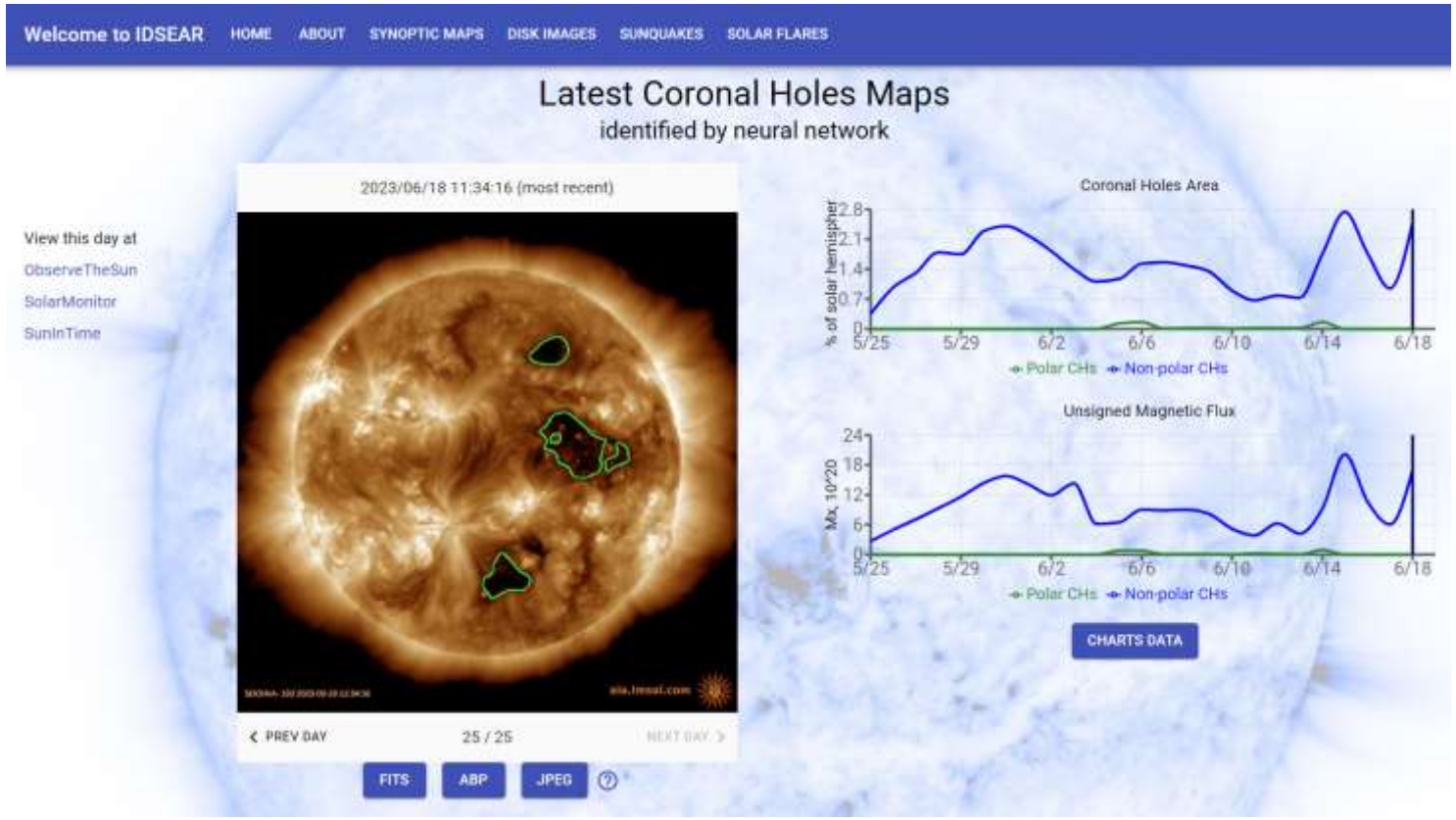


Fitted models reproduce observational data

Illarionov et al. Neural differential equations for the solar dynamo. Physics Review E. 2026

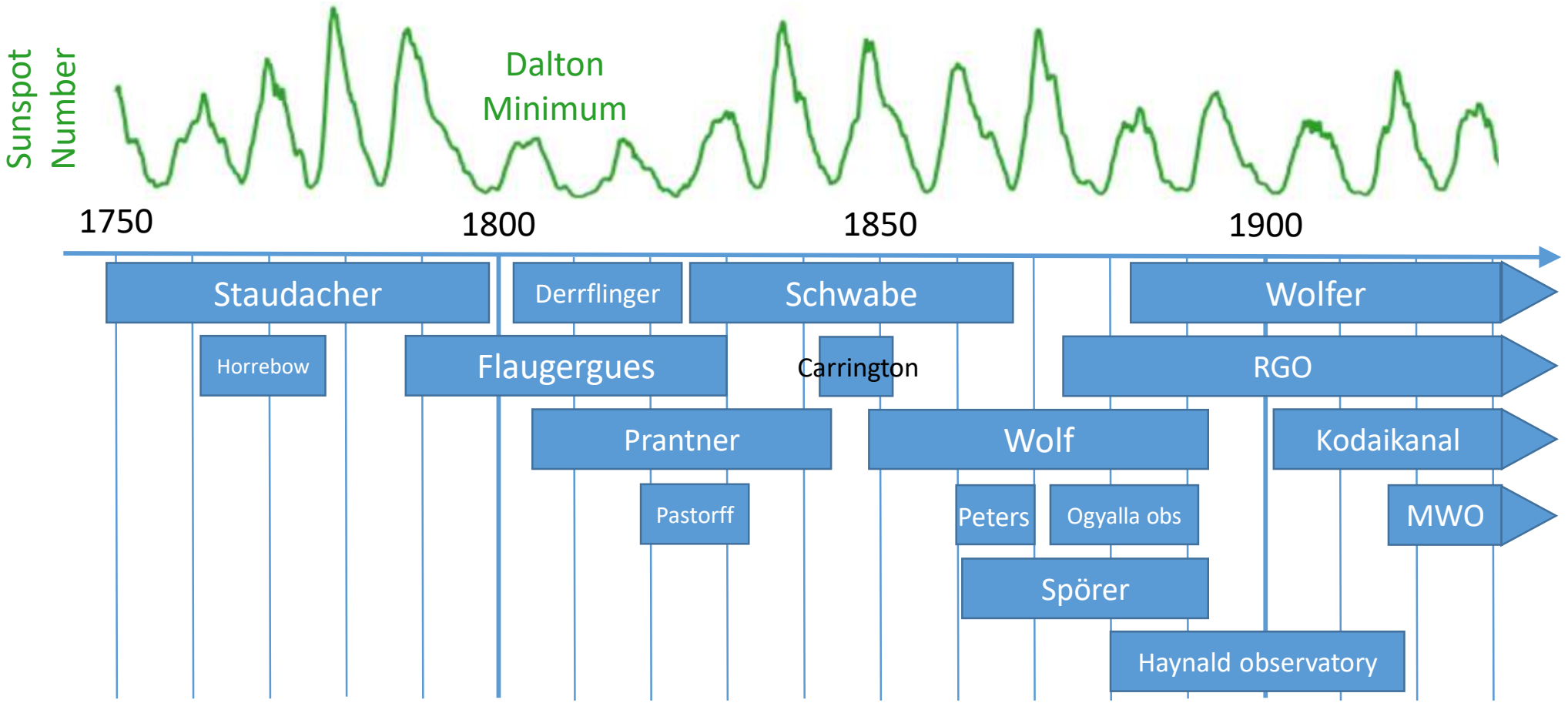
Automatic segmentation of images

Modern telescopes capture the Sun each 12 seconds. Without AI, we can not benefit from such datasets

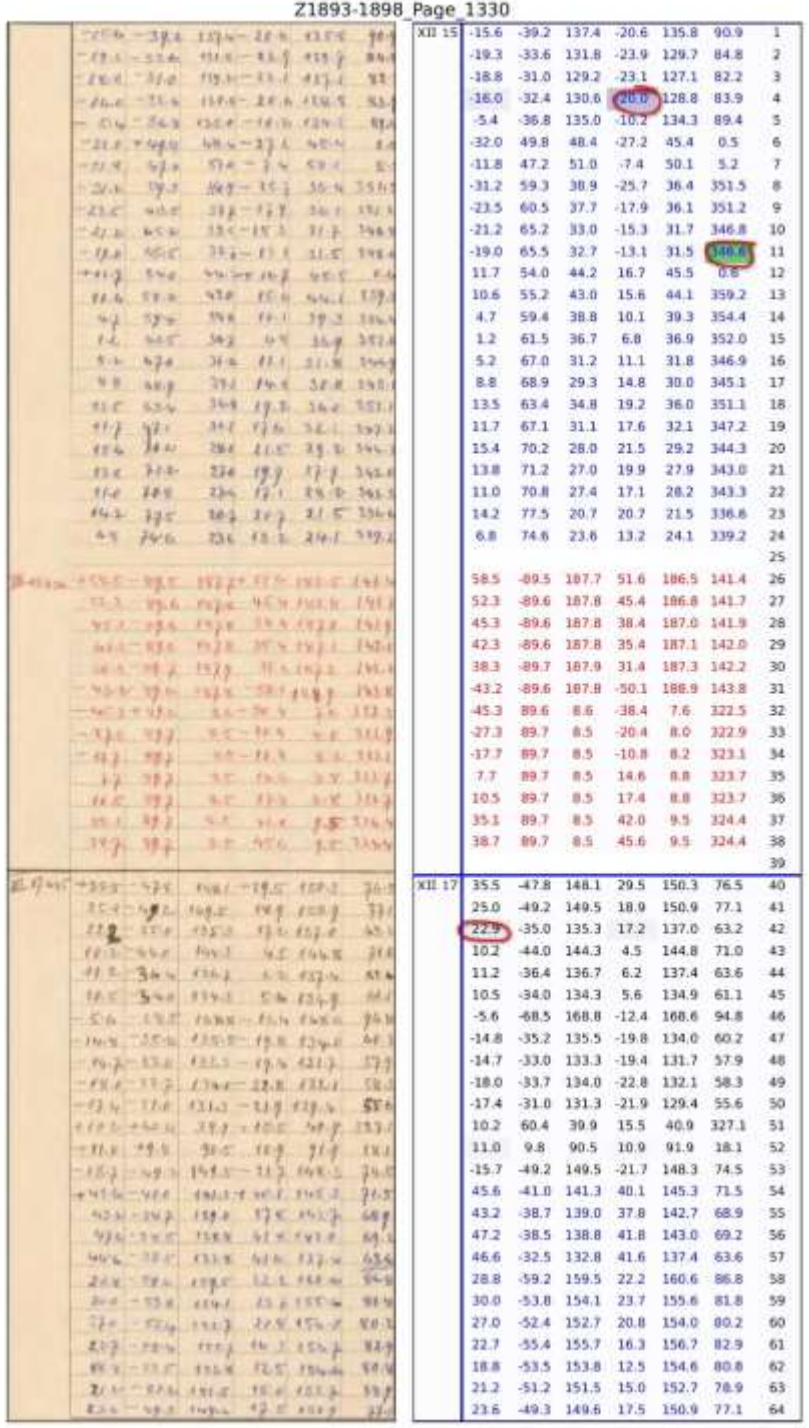


Illarionov et al. Machine-learning approach to identification of coronal holes in solar disk images and synoptic maps. Astrophysical Journal. 2020

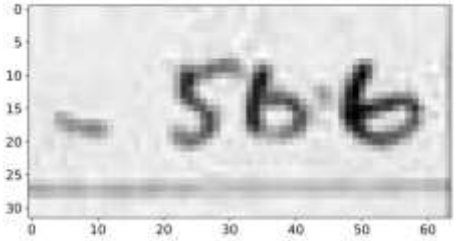
Historical archives on solar activity



Catalogs of the Zurich observatory
10000+ pages of handwritten tables



Digitization of handwritten archives



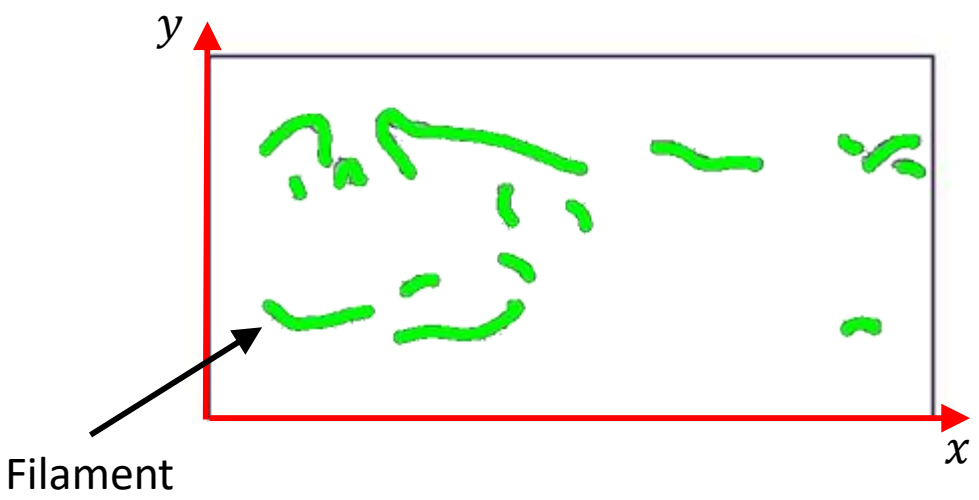
CRNN →

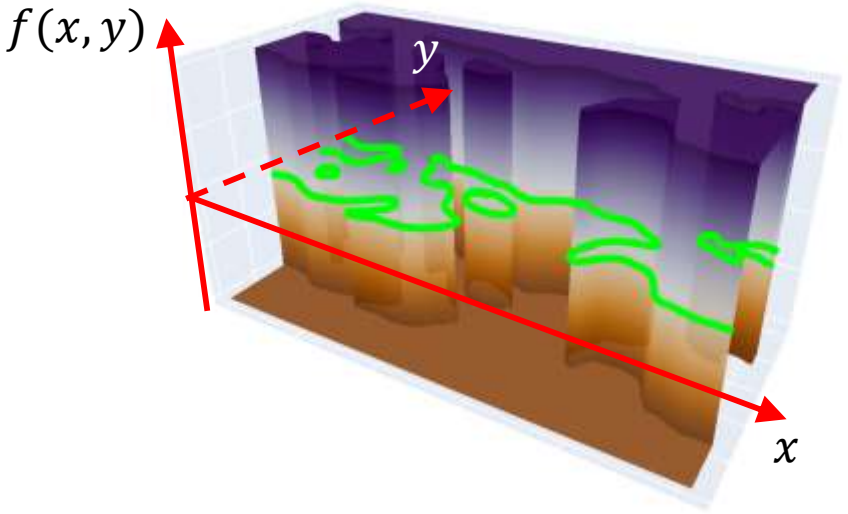
Number	Confidence
-56.6	0.89
-56.0	0.08
56.6	0.01
-56.2	0.01
-50.6	0.01

Conventional supervised approaches worked poor, real-time self-supervised model helped to get exceptional accuracy and confidence

Illarionov & Arlt. Reconstruction of the Solar Activity from the Catalogs of the Zurich Observatory. Solar physics. 2022

Reconstruction of magnetic-field polarity





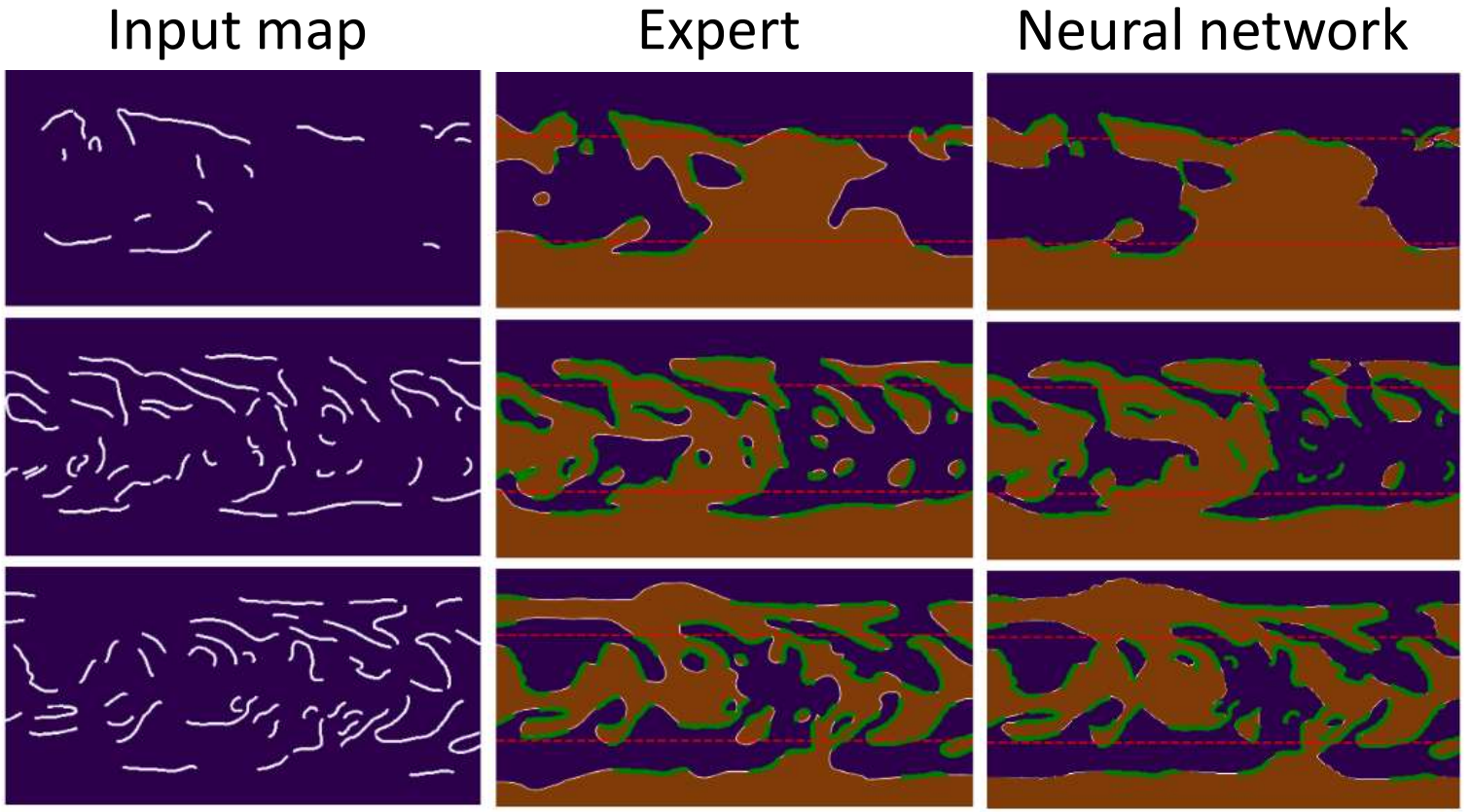
Filaments indicate position of the polarity inversion line in the (x, y)-plane

We reconstructed a function $f(x, y)$ that changes sign crossing the polarity line

$$\mathcal{L}(f) = w_1 \left| \int_{\mathcal{C}} f \, dV \right| + w_2 \int_{\mathcal{F}} f^2 \, dV - w_3 \int_{\mathcal{C} \setminus \mathcal{F}} |f| \, dV$$

Loss function used to optimize the function $f(x, y)$

Reconstruction of magnetic-field polarity



Kisielius & Illarionov. Machine Learning for Reconstruction of Polarity Inversion Lines from Solar Filaments. Solar Physics, 2024.



Conclusions

In the presence of intermittency, numerical modeling can misrepresent dynamical system behavior. Analytical insights are essential to ensure the validity of numerical results

A data-driven approach and ML models effectively minimize uncertainties in theoretical frameworks

The current digital transformation of science accelerates the transition toward autonomous research